What an Auxiliary Cue Must Carry: LiDAR Body Pose for Pedestrian Trajectory Forecasting

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***Abstract*—Gait and facing appear to encode pedestrian intent that a position history lacks, making body pose an attractive forecasting input. On the Waymo Open Motion Dataset (WOMD), with SMPL poses recovered from LiDAR, we find that whether it pays depends on the track it is added to. The decisive comparison is correct pose against the same architecture fed *another pedestrian's* pose, which separates body information from an extra branch's capacity. On WOMD's tracks it returns $+0.04\%$ (p=0.93) at our largest scale, and the data bound the content effect to $\pm 1.7\%$. Those tracks are offboard auto-labels, refined with future frames. Replace the observed history with what a causal onboard tracker would report, and own pose beats a stranger's by 6.0% (p=0.004) at 30 cm of detection noise, on moving pedestrians, whose gait carries the speed the tracker loses. Degrading the box heading instead leaves the null intact, because body facing is a noisier copy of that heading (12.3° against 3.9°). An oracle through the same branch converts the agent's own future bearing at every accuracy we trained, so the pathway is not the obstacle. Pose is redundant with an offboard track, not with a causal one. Before integrating a modality, measure what it carries beyond the inputs the model already has, and measure it against the inputs a vehicle will have.**

***Index Terms*—pedestrian trajectory prediction, human pose estimation, LiDAR, autonomous driving, offboard labels**

I. Introduction

Predicting where a pedestrian will go is a safety-critical function of an automated vehicle, and body pose is an obvious candidate input. A person's stance, gait phase and facing direction plausibly reveal intent before their position changes, and modern perception stacks already produce human mesh or keypoint estimates. Several recent systems condition trajectory forecasts on pose [1]–[4].

This paper reports that, on the Waymo Open Motion Dataset (WOMD), pose conditioning does not measurably improve pedestrian trajectory accuracy, and that the reason is the track it is added to. WOMD's boxes are offboard auto-labels, refined with the whole track, future frames included [5], [6]. Against that track the body restates what the model already knows. Against the history a causal onboard tracker would report, the same body pays. The reason can be checked quantitatively, and cheaply, before a modality is integrated at all.

We make four contributions.

A controlled null on offboard tracks. We measure pose conditioning against a shuffled-pose control: the same architecture receiving a different pedestrian's body motion. This isolates information content from the capacity cost of an extra branch, and it is the comparison a practitioner should demand before adopting any auxiliary modality. The control is null at three training scales and under two integration styles. It is also null across the three pose encodings we run it against on the corrected export (6D joint rotations, agent-frame root, and eight frame-invariant descriptors), with constant-pose as a capacity control. Two further encodings (3D joint positions and a root-orientation-free variant) and a GRU encoder predate the export fix of Sec. III and were run against no-pose rather than against a stranger's pose, so we neither count them as controlled nulls nor quote numbers from them. An oracle arm, which replaces the body with a signal encoding the agent's own future, improves minADE by 34.8% through the same branch, so the null is a property of what pose adds to these tracks, not of the plumbing.

Where it pays. We replace WOMD's observed history with what a causal tracker would report, per-frame detection noise filtered by a forward-only Kalman filter, and repeat the content test. Pose now pays: own pose beats a stranger's by 6.0% at 30 cm of detection noise, on moving pedestrians, whose gait carries the speed the tracker loses. Corrupting the box heading instead leaves the null intact. The body carries gait that an offboard track makes redundant and a causal one does not.

What the pathway converts, and what it does not. We corrupt that oracle with a controlled angular error to ask what kind of information the branch turns into accuracy. Every accuracy we trained converts, including one uniformly worse than an input the model already has, so the ladder does not set an accuracy a cue must reach. It separates a precision term with a 3.0° half-life from a slowly decaying term that is future-side mode selection. Body facing supplies neither: it is a noisier copy of the box heading where pedestrians move, and reaches 41.2° where they do not, which is exactly the case velocity cannot cover. We also show the same ladder run at 579 scenes reports a clean accuracy threshold that 25k scenes removes, so an underpowered ladder manufactures a

floor where there is none.

A measurement caveat. Aggregate MPJPE cannot distinguish a pose estimator that tracks a joint imprecisely from one that ignores the input and emits its prior. Scoring each joint against an always-predict-the-mean-pose baseline, MPJPE is uninformative about which joints are actually recovered ($\rho = -0.11$ across joints, n.s.) and inverts the ordering for the head. Accepting a pose estimator into an AV stack on a single error number risks importing a prior.

We also report, as a cautionary note, that our first attempt at this study was invalidated by an export bug that silently substituted other pedestrians' bodies, a failure any pose pipeline keyed by object identity can reproduce.

II. Related Work

Trajectory forecasting on WOMD is dominated by anchor- and query-based transformers. We build on MTR [7], [8], which is representative and strong. Human mesh recovery from LiDAR is maturing [9], with motion priors used to complete sparse observations [10]–[12]. Intent-focused work on pedestrian datasets [13], [14] targets crossing decisions rather than metric trajectories. Kress et al. [4] report pose-based gains for vulnerable road users from a research vehicle, at close range and with camera-derived pose. Methodologically our closest precedent is Makansi et al. [15], who find by feature attribution that trajectory predictors draw almost nothing from an input the field had assumed was central.

Relation to prior positive results. Pose conditioning has reported real gains, and our mechanism predicts where they come from rather than contradicting them. Social-Transmotion [1] lowers ADE on the synthetic JTA dataset from 0.99 to 0.89 with *ground-truth* 3D pose, and, in the harder case for us, from 0.40 to 0.36 on JRDB with *estimated* pose. Social-Pose [2] reports about 25% on JTA ($1.20 \rightarrow 0.90$) and on JRDB. Salzmann et al. [3] report $\approx 10\%$ from onboard-estimated keypoints at ranges up to 7 m, specifically for agents with little position history. A short history is a poor track, the condition under which we find pose paying (Sec. IV-E).

Le et al. [16] reach the opposite conclusion on JRDB and, in doing so, say the most useful thing about ours: they localise the gain to *lower-body* 3D keypoints (13% ADE, a further 1–4% from derived biomechanical cues). Their cue is our weakest channel. Of the twelve limb joints in Table V the knees and ankles that carry gait sit at the bottom of the range (SKILL $+0.46$ to $+0.49$, against $+0.63$ and $+0.64$ for the elbows), and what leg motion we do recover is damped: predicted knee and ankle spread is 84–91% of ground truth, where elbows and wrists reach 96–99%. A panoramic camera at a few metres resolves a swinging leg. A few dozen LiDAR returns at 40 m recover its mean position and lose much of its amplitude. So the two results do not conflict: theirs is what this cue is worth when the limbs are resolved and the predictor has no heading input, ours what it is worth when the trajectory already carries the body's direction and speed. Degrade our track to a causal tracker's and a gain appears on moving pedestrians, where gait acts (Sec. IV-E). Sparsity alone does not account for the difference, since our content test is null even on the best-resolved agents (Sec. IV-D).

The most informative comparison is within Social-Pose itself. Alongside JTA it evaluates on Pedestrians and Cyclists in Road Traffic, the one vehicle-captured dataset there, and, with *ground-truth* 3D pose, ADE moves from 0.58 to 0.57 m, by 0.01 m or less for all five architectures tested, against the $\approx 25\%$ the same encoder delivers on JTA. The authors attribute this to the shorter horizon and the absence of interacting neighbours. Our measurements suggest an additional reason that travels with the sensor rather than the horizon. Either way, the closest published setting to ours already reports approximately our result.

Two properties separate the remaining positive settings from an AV stack, and both follow from the mechanism in Sec. V. *Range and sensor:* JRDB is an indoor and campus robot dataset in which pedestrians are metres away and pose is camera-derived, whereas WOMD pedestrians are observed from 2 to 75 m and often return a few dozen LiDAR points. *The baseline:* those predictors consume positions, not a box heading, so body facing is new information there and partly duplicated here. On controls, none reports a shuffled-pose arm, though the confound differs by design. Social-Transmotion masks modalities within one network, so no extra parameters are involved there, whereas a dedicated pose branch pays the capacity cost we measure ($+1.7\%$ minADE), the same order as the smaller reported gains.

III. Setup

A. Data and a data-integrity failure

We use WOMD [5] with SMPL [17] poses recovered from LiDAR by the Waymo-3DSkelMo pipeline [18], which fits per-frame bodies with LiDAR-HMR [9] and completes them using AMASS-trained motion priors [10]–[12]. Poses cover the 11 observed frames, and the prediction horizon is 8 s at 10 Hz. The branch is not starved: every predicted agent carries at least one valid pose frame, and 86.7% of observed frames carry one, so sparsity of pose is not an explanation for anything that follows.

One property of that pipeline bears on Sec. V: when a pedestrian is moving and the raw LiDAR estimate of root rotation deviates from the past-trajectory direction by more than $90°$, the pipeline snaps the root to the direction of travel. Body facing is therefore partly trajectory-derived for moving pedestrians. This reinforces the redundancy we report rather than weakening it, and it does not reach the stationary-onset analysis, whose events are all below 0.3 m/s.

Our first pass at this study was invalid. The export keyed pose clips by a scene-*local* object identifier, so clips collided across scenes and nearly every training sequence carried another pedestrian's body. A per-scene re-export (299,220 scenes, 1.81M sequences) cut the median error between root heading and travel direction from $41.5°$ to $11.4°$ and raised the gait–speed correlation from $r = 0.26$ to 0.58. We report it because the failure is silent (training converges, losses

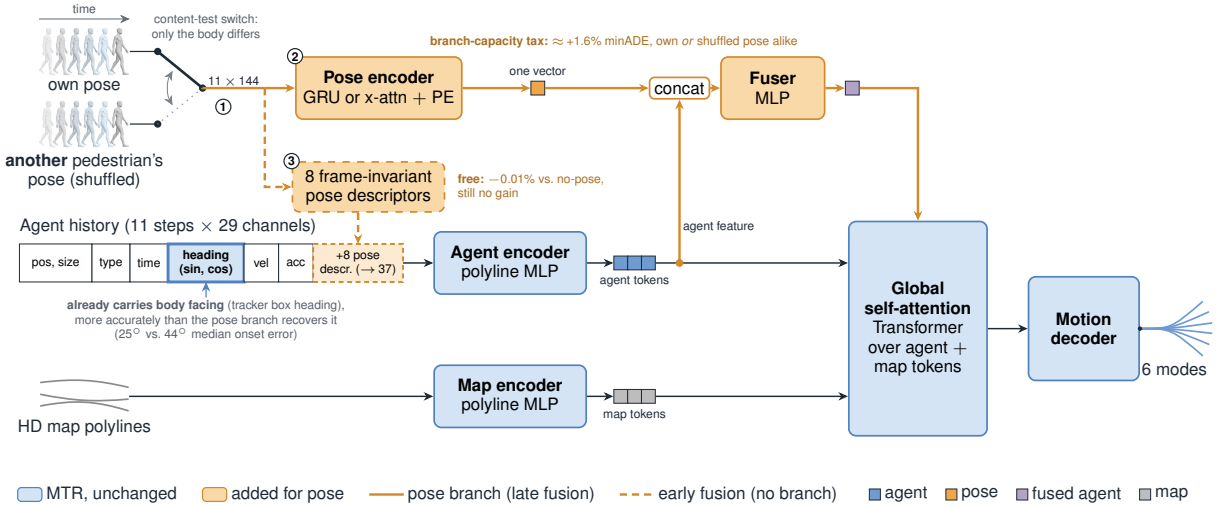


Fig. 1. The two arms differ only in *whose* body enters the model. The content-test switch (1) feeds the predictor either the pedestrian's own SMPL sequence or another pedestrian's, holding parameters, optimiser and input dimensionality fixed. Everything downstream is shared. Blue: MTR, unchanged. Orange: added for pose. The agent-history strip shows that the labelled box heading is *already* an input. Solid orange (2): the pose branch (late fusion), which costs $\approx$1.6% minADE whether it carries the right body or the wrong one. Dashed (3): early fusion, which removes that cost, still yields no benefit from pose, and converts an uncorrupted oracle on every seed at 25k scenes (Sec. V-C).

look normal) and identity-keyed joins are common in pose pipelines. Every number here is from the corrected export.

B. Arms and the content test

Three arms share one architecture (Fig. 1). *No-pose* receives trajectories and map. *Pose* adds the pedestrian's own SMPL sequence. *Shuffled* receives another pedestrian's sequence, with identical parameter count, optimiser and input dimensionality. A fourth arm, *oracle*, replaces the SMPL features with a six-channel encoding of the agent's own future displacement at 3 and 8 s, leaving the branch and its masks untouched. It shows the pathway can carry information at all (Sec. IV-C).

Pose versus no-pose confounds information with branch capacity, while pose versus shuffled isolates content. We call the latter the *content test* and treat it as decisive. Relative differences are $(A - B)/B$, so negative favours A.

Inference is over *seeds*, which is the unit the treatment is applied to: every comparison between trained arms is a paired t-test on per-seed differences. The two exceptions are flagged where they occur: per-pedestrian subpopulation tests and the per-joint intervals of Table V, which resample pedestrians. We also report a hierarchical bootstrap that resamples seeds and then pedestrians, and it agrees. We do not quote a percentile bootstrap p-value, because with five seeds whose differences share a sign it returns the resolution floor $1/N_{\text{boot}}$ rather than a calibrated tail probability, and a pedestrian-only bootstrap ignores between-seed variance entirely.

C. Protocol

Every arm trains for 30 epochs with AdamW (batch 2 per GPU, weight decay 0.01, gradient-norm clipping 1.0) under a step schedule that halves the learning rate at epochs 10, 20 and 25, and the decoder predicts K=6 modes. The learning rate is chosen per rung from a grid run separately for each arm: 1×10^{-4} at 579 scenes, 5×10^{-5} at 5k and 2.5×10^{-5} at 25k, where every arm peaks at the same interior optimum. Validation is fixed at 5,171 scenes and 16,486 pedestrians. Headline numbers are taken at each run's best validation epoch over a fixed evaluation schedule, and Sec. IV-B reports what changes under a fixed-epoch rule. Official WOMD mAP and gated miss rate need the full retained prediction set, which our per-epoch evaluation does not store, so we report minADE, minFDE and miss rate at 2 m. Code, configs, per-seed metrics and the analysis scripts are released with the paper.

IV. Results

A. The null on offboard tracks, and its bound

Table I trains all three arms at three scales. Data alone improves the baseline by 19.2%, while the content test is null at every rung, in minADE and in minFDE alike, and in miss rate at 2 m, where the content test is +2.3% at 25k (p=0.21) and +4.5% fine-tuned (p=0.19): null, and with its sign against pose on all three metrics. The rungs are not equally sensitive, and we state what each one could have seen: at 80% power the minimum detectable content effect is 3.5% at 579 scenes (n=5), 7.6% at 5k (n=3) and 2.2% at 25k (n=5). The 5k rung is therefore weak evidence on its own and we do not count it as an independent null. The weight falls on the 25k rung, which is both the largest scale and the most sensitive. At 25k the paired seed-level interval on the content difference is $[-1.59\%, +1.70\%]$, so at the largest scale we train the data exclude a content effect beyond roughly 1.7% in either direction, a bound tighter than the 1.6% cost of the branch that would carry it. The pedestrian-resampling bootstrap gives a visibly narrower interval ($[-0.0040, +0.0047]$ m, $\approx\pm1.2\%$). With five clusters it undercovers, so we quote the wider seed-level bound throughout.

TABLE I

CONTENT TEST ACROSS TRAINING SCALE, VALIDATION FIXED. THE LEARNING RATE IS SELECTED PER RUNG BY A GRID RUN SEPARATELY FOR EVERY ARM, AND ALL ARMS PEAK AT THE SAME VALUE (SEC. III-C). PERCENTAGES ARE DIFFERENCES OF ARM MEANS, AND p-VALUES ARE PAIRED t-TESTS OVER PER-SEED DIFFERENCES. NEGATIVE FAVOURS POSE.

Training scenes	579	5k	25k
Pedestrians	6,348	31,302	142,807
Seeds	5	3	5
minADE (m)			
No-pose	0.4599	0.4176	0.3716
Pose	0.4620	0.4138	0.3778
Shuffled	0.4589	0.4181	0.3777
Pose vs. no-pose	$+0.45\%$	-0.90%	$+1.66\%^{*}$
Content test	$+0.66\%$	-1.02%	$+0.04\%$
p (content)	0.52	0.56	0.93
minFDE (m)			
No-pose	0.9922	0.8992	0.8049
Pose	0.9824	0.8875	0.8131
Shuffled	0.9780	0.8900	0.8104
Pose vs. no-pose	-0.99%	-1.30%	$+1.02\%^{\dagger}$
Content test	$+0.45\%$	-0.28%	$+0.33\%$
p (content)	0.63	0.83	0.61

$^{*}t(4)=10.3$, $p=5\times10^{-4}$. The shuffled arm pays the same cost, $+1.63\%$, $t(4)=2.8$, $p=0.048$.
$^{\dagger}t(4)=6.1$, $p=0.004$.

TABLE II

CLEAN CORE AT 579 SCENES, $n=5$ SEEDS, BOTH REGIMES. THE SHUFFLED ARM HAS THE LOWEST MEAN IN BOTH: ANOTHER PEDESTRIAN'S BODY IS NOMINALLY THE BEST INPUT.

Arm	Per-seed minADE	Mean $\pm$ sd	Content
From scratch			
No-pose	.458 .443 .453 .495 .450	0.4599 ± 0.0206	—
Shuffled	.454 .467 .451 .455 .467	0.4589 ± 0.0075	—
Pose	.471 .456 .453 .457 .474	0.4620 ± 0.0094	$+0.66\%$
Fine-tuned on a full-WOMD backbone			
No-pose	.372 .366 .367 .373 .368	0.3692 ± 0.0031	—
Shuffled	.364 .363 .364 .374 .373	0.3676 ± 0.0055	—
Pose	.364 .372 .371 .374 .373	0.3708 ± 0.0040	$+0.88\%$

After pretraining a trajectory-only backbone on full WOMD and fine-tuning ($n=5$), the content test is $+0.88\%$ ($p=0.18$), again with its sign against pose (Table II).

Seed count decides what Table II appears to say. On the first three seeds the same runs read as *pose harms* from scratch ($+1.85\%$) and *shuffled is best* after fine-tuning (-1.28%). At $n=5$ both shrink roughly fourfold. Both three-seed effects are significant under a pedestrian-resampling bootstrap ($p=0.037$, $p=0.016$) and neither is under a paired t over the same seeds ($p=0.18$, $p=0.12$): three seeds do not resolve a 1% effect here, and a bootstrap over the wrong unit will not say so.

B. Two effects that are real, and are not pose

A branch-capacity tax. At 25k both pose arms are worse than no-pose ($+1.66\%$ at $p=5\times10^{-4}$ and $+1.63\%$ at $p=0.048$) and indistinguishable from each other. An extra input branch costs $\sim1.6\%$ whether it carries the right body or the wrong one.

This is an operating-point statement, and the operating point matters: it is measured at each run's best validation epoch. Models with wider inputs overfit sooner, so the sign of a pose-versus-no-pose comparison is not stable under a fixed late-epoch rule, where the no-pose arm has degraded further by then ($+8.0\%$ in favour of pose at epoch 30 with $n=3$, and $+0.9\%$ at $n=8$, both n.s. and neither resolved at these seed counts). What is stable is that the comparison moves with the stopping rule, so we never compare architecturally different arms at a fixed late epoch. The *content* test is null under either rule: its two arms have identical input dimensionality, so this asymmetry does not arise. Replacing the 144-d branch with eight frame-invariant descriptors fused into the existing agent channels removes the cost (-0.01% versus no-pose at $p=0.99$, and -1.67% versus the branch at $p=0.12$, $n=3$) without creating a benefit: pose can be carried for free, and still does not pay. That freedom is real, and Sec. V-C is what establishes it: fed an uncorrupted oracle at 25k scenes this pathway converts it on every seed, so a channel it costs nothing to carry is still a channel it reads. We quote the seed-level p here for consistency with the rest of the paper. A pedestrian-resampling bootstrap makes the branch contrast look decisive ($p=0.006$), which is exactly the unit error Sec. III warns against.

Tuning dominates the treatment. At 5k the learning rate inherited from the smaller rung made pose look catastrophic ($+37\%$). With per-arm tuning it reads -0.90%, n.s. Across the 25k grid the learning rate moves minADE by up to 16.7% against a content signal of $\sim0.05\%$: a pose-versus-baseline comparison that does not tune each arm separately is a claim about the learning rate.

C. The branch is not inert

A null measured through one pathway invites the obvious objection that the pathway is dead. The shuffled control does not dispose of it: it shows the model cannot distinguish its own body from a stranger's, which is equally consistent with a branch that transmits nothing at all. The branch-capacity tax shows the branch perturbs the output, not that it carries signal, and the fusion residual is zero-initialised, so a dead branch and an uninformative one begin at the same place.

We therefore ran the positive control. The *oracle* arm replaces the SMPL features with a six-channel encoding of the agent's own future displacement at 3 and 8 s, which is genuinely predictive by construction. It keeps the branch, its parameter count, its input dimensionality and its per-frame validity masks exactly as the pose arm has them, so the oracle is visible only where a real pose would have been.

Over three seeds the oracle arm reaches 0.2945 against no-pose's 0.4514 on the same seeds, a change of -34.8% ($t(2)=-81.4$, $p=1.5\times10^{-4}$), and -35.6% against the shuffled arm. The oracle is of course not a usable model, since it reads the future. Its only job is to calibrate the pathway. It does so emphatically: the branch converts information into accuracy at roughly twenty times the magnitude of the bound our content test places on pose. The null is therefore a statement about

TABLE III
ONBOARD-QUALITY INPUTS AT 25K SCENES, n=3 SEEDS PER ARM. EACH CONDITION DEGRADES ONE INPUT OF THE BASELINE AND REPEATS THE CONTENT TEST. LOSS IS THE NO-POSE BASELINE AGAINST ITS CLEAN CELL, AND NEGATIVE FAVOURS POSE.

Degraded input	Loss	Content	p	vs. no-pose
None	—	$+0.41\%$	0.60	$+1.97\%$
Box heading, fixed error $\mathcal{N}(0, \sigma^2)$				
σ=15°	$+5.8\%$	-0.14%	0.90	-0.39%
σ=30°	$+12.6\%$	-0.11%	0.93	$+1.34\%$
Track, causal tracker on detection noise s				
s=10 cm	$+14.1\%$	-2.14%	0.06	-1.92%
s=30 cm	$+32.3\%$	$\mathbf{-6.05\%}$	**0.004**	-4.60%

what SMPL pose contains, not about whether the architecture can use what it is given.

D. Where a benefit could still hide

On an eight-seed replication of the content test at 30 epochs, scored on a held-out half of validation (6,764 agents), it is null at every horizon and least favourable at the shortest ($+0.48\%$ at 2 s, -0.04% at 8 s). It is null in all 20 tests across quartiles and top deciles of constant-velocity residual, heading variation, speed range and peak acceleration. It is also null for standing-then-moving pedestrians, the subpopulation built to favour it (n=1,809 pedestrians, $+0.38\%$, epoch-30 checkpoints on an expanded validation split, p=0.80 by a pedestrian-resampling bootstrap). That is the one place we quote a bootstrap, because this subpopulation is defined per pedestrian.

It is also null where the body is best measured, which is where a pose-quality explanation predicts an effect should appear. Stratifying the 25k content test by LiDAR return count gives -0.37% ($[-1.87, +1.14]$) on the 2,372 agents with at least 200 returns and -0.08% on the 10,902 with at least 30, and stratifying by range gives -0.14% ($[-1.61, +1.33]$) inside 10 m. Nine strata, no effect, at n=5 throughout. Whatever the null is, it is not only sparsity at range.

Across the roughly forty content tests this study reports outside the onboard-track ladder of Sec. IV-E, two reach $p<0.05$, and both favour the stranger. In the stationary subpopulation of the heading ablation, correct pose is $+1.8\%$ *worse* than a stranger's (n=8, p=0.011), and in the stationary subset of the onboard-heading ladder at σ=30° it is $+1.3\%$ worse (n=3, p=0.049, Sec. V-B). At that multiplicity one or two such results are expected by chance, and their direction is against a pose mechanism rather than for one. We report them so that the released metrics contain no surprise. The tests of Sec. IV-E that favour pose are a different kind of result. They test one hypothesis fixed before the runs, grow with the noise and agree in sign on every seed.

E. Where it pays: an onboard-quality track

Everything above is measured on WOMD's tracks, and those are offboard auto-labels. An offline network refines every box with the whole track, future frames included [5], [6], and the velocity the baseline receives is exactly the backward difference of those refined positions at all 954,751 pedestrian time steps we checked. The observed history is as clean as the labeller could make it. A deployed stack has a causal tracker instead, so we simulate one. Each observed position receives detection noise $\mathcal{N}(0, s^2)$ per axis, drawn once rather than per epoch, and a forward-only constant-velocity Kalman filter, started at rest, turns the detections into the positions and velocities the model sees. The box heading and the future targets are untouched, which keeps this apart from the heading corruption of Sec. V-B. At prediction time the tracker is off by 7.5 cm and 0.23 m/s in median at s=10 cm, and by 21 cm and 0.46 m/s at 30 cm, against a median pedestrian speed of 1.1 m/s.

The null breaks (Table III). At 30 cm own pose beats a stranger's by 6.05% (p=0.004, 95% CI $[-7.7, -4.4]$) and beats no pose by 4.60% (p=0.05). Every seed agrees (-6.4, -5.3 and -6.5%), and so does every fixed late epoch (-6.5 to -6.6% at epochs 26, 28 and 30), so this is not the stopping rule. At 10 cm the effect is a third as large, -2.14% against shuffled (p=0.06) and -1.92% against no pose (p=0.03). It sits where gait would put it. On the 8,757 pedestrians walking at 0.8 to 1.6 m/s it is -3.8% (p=0.02) at 10 cm and -7.3% (p=0.003) at 30 cm, while on the 3,336 standing at prediction time it is significant at neither. The box heading still states a walker's direction, so what the tracker spoils is speed and path, and a stride carries speed: gait and speed correlate at r=0.58 on the corrected export (Sec. III).

The clean track cannot reach the model through the body. The pose stream holds joint rotations only, with no translation, covers only the 11 observed frames, and the pose and shuffled arms receive identical tracks. Two caveats bound the result. The tracker is simulated, with Gaussian detection noise rather than a detector's correlated, occlusion-driven errors. And the bodies are still fitted inside offboard boxes, so an onboard pose estimate would degrade too and these gains are optimistic for this estimator. What they establish is the direction: the same body that adds nothing to an offboard track adds measurably to a causal one.

V. WHY POSE DOES NOT PAY ON OFFBOARD TRACKS

A. What the oracle ladder measures

The content test bounds what pose is worth, and the oracle of Sec. IV-C shows the branch can carry information. Neither says what *kind* of information this pathway turns into accuracy. We measured that by degrading the oracle.

Each rung of Table IV feeds the branch the same direction-only oracle, the bearing of the agent's own future displacement at 3 and 8 s, corrupted by a fixed angular error. Each is labelled by the median bearing error actually delivered: 0.6745σ by construction, since the corruption is a fixed per-agent rotation, and measured to within 0.7° of it on 11,328 scored agents. Displacement magnitude is withheld, so the channel carries direction alone. The error is drawn once per pedestrian, not per epoch: a real estimator's error is a property of the observation, and resampling it would let training average it away.

At 25k scenes every rung converts, and the gain stays large: -34.50% with an exact bearing, -19.77% at 5.0°, -15.68%

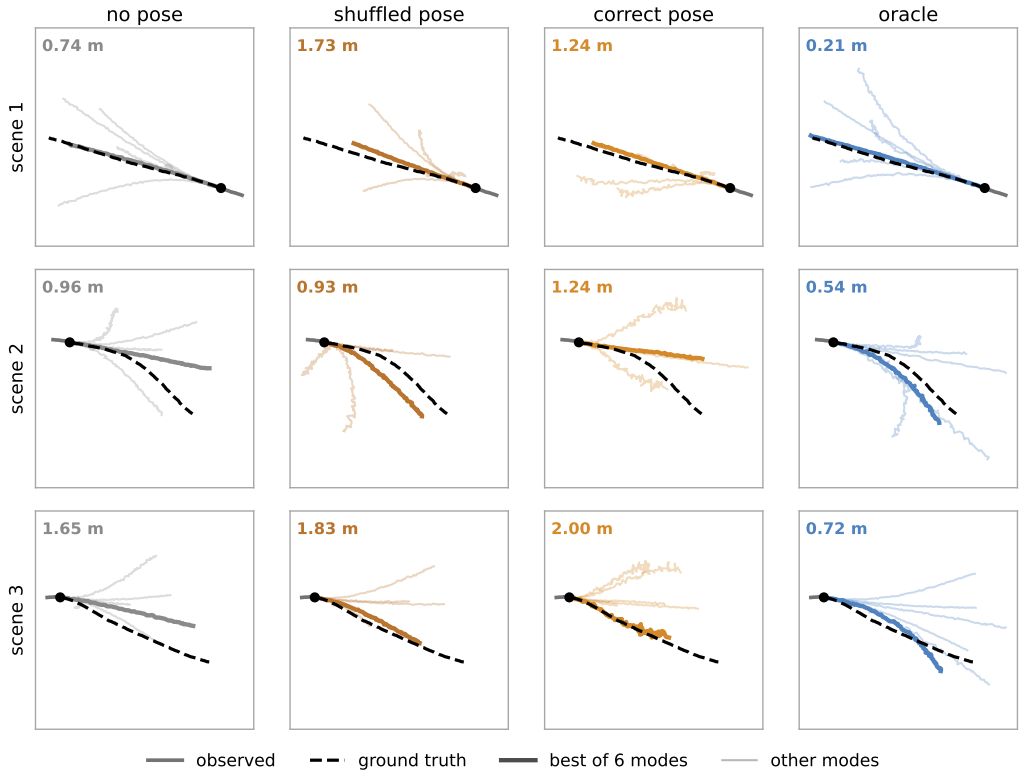


Fig. 2. Qualitative comparison of the four arms on the same three agents (seed 101, epoch 30, the epoch for which full trajectory dumps were retained). Scenes are picked by percentile of the *no-pose* arm's minADE (60th, 76th, 92nd) among agents with a full 8 s future and more than 3 m of travel. Selection is on the baseline alone, never on the treatment. Thin lines are the six predicted modes, thick the best of them, dashed the ground truth, grey the observed past. Fed the agent's own future bearing, the oracle bends its modes onto the true path through the *same* branch that carries pose, while correct pose is not consistently distinguishable from a stranger's pose or from no pose. Per-panel values are single-seed illustrations: across seeds the oracle's advantage is stable (-34 to -40%), whereas the content difference at this fixed epoch is not (-16.2, -8.0, -0.2% for seeds 101/202/303) and is null at the best-epoch operating point used for inference (Table I).

at $10.0°$ and -13.57% at $15.1°$, all at $p \leq 0.0016$ against minimum detectable effects of 1.4–2.9 pp. The decay is not exponential: one exponential fitted to the four rungs leaves residuals (measured minus fitted) of -1.5, $+3.2$, $+0.3$ and -2.5 pp, a systematic pattern in θ against those MDEs rather than scatter about the curve. The curve decelerates, and two terms fit it to 0.4 pp:

$$\Delta minADE(\theta) \approx -\left[13.1\% + 21.4\% \cdot e^{-\theta/4.4°}\right], \quad (1)$$

a precision term with a half-life of $3.0°$ over a term the corruptions in this ladder do not resolve. That term is slow, not permanent: it is flat to $15.1°$ here, while the same ladder at 579 scenes reads -0.1 to $+2.7\%$ from $24.9°$ outward, with intervals that exclude the $\approx 6\%$ a persistent term scaled from that ladder's own $\theta{=}0$ amplitude would predict (Table IV). It must decay somewhere beyond our rungs, since a $60°$ median error is close to uniform on the circle. The seeds agree closely: 12.3–14.2% persistent, amplitudes 20.8–21.8%, half-lives 2.8–3.4°.

The angular axis is not an acceptance threshold. It is tempting to read θ as the accuracy a cue must reach: feed the branch a channel whose bearing error is θ, expect the gain at rung θ. The ladder refutes that reading itself. The baseline already receives the labelled box heading, as a per-timestep sine–cosine pair for every agent (Fig. 1), and that heading estimates the *same* future displacement bearing at $3.9°$ in median, better than the $15.1°$ rung's channel at every percentile ($9.3°$ against $25.8°$ at the third quartile, $20.1°$ against $36.8°$ at the ninetieth). A channel uniformly less accurate than an input the model already has still buys -13.6%. Nor does the branch pay for merely being occupied: an extra branch carrying uninformative content *costs* $\sim 1.6\%$ (Sec. IV-B), and the same branch fed another pedestrian's pose returns $+0.04\%$ (Table I). The 34.5% is content, and the only content this channel has is the realised future.

The surviving term is future-side mode selection. Eq. (1) splits the gain, and the two terms behave differently in ways that identify them. Per-agent records at each run's own best epoch, paired between arms on 16,486 agents and averaged over seeds, put the persistent term at the end of the trajectory: refitting the decomposition at each horizon leaves the precision half-life at 2.9–3.9° throughout, while the persistent share of the gain rises from 21% of the 2 s error reduction to 39% at 8 s and 53% at the endpoint. The selection reading predicts exactly that, and it predicts more: corrupting the bearing should barely

affect *which* of the six modes is right. It does not. The gap between the top-ranked mode and the best of six, which is what the forecaster loses by choosing wrongly, falls from 0.67 m to 0.26 m with an exact oracle, and still to 0.29 m at 15.1°: the corrupted channel closes 92% of the perfect oracle's selection gap while recovering 39% of its minADE. The absolute gain is also largest in the quartile where the baseline's own modes disagree most at the endpoint (0.150 m against 0.111–0.130 m elsewhere, at θ=0). A coarse future bearing says which way the agent went, and only a precise one refines where along that way, and that refinement is what 3° of error destroys.

minADE takes the best of six modes whatever their rank, so selection reaches it only through the decoder's scoring step, which keeps six of 64 intention queries by predicted likelihood: a better ranking changes which six survive. That is also where a pose effect could hide from our headline metric, so we scored the ranking columns directly on the 25k arms. They are null as well: against a stranger's pose, top-1 ADE is −2.07% (p=0.11, MDE 3.7 pp), best-mode rank −0.93% (p=0.22) and endpoint mode spread +0.17% (p=0.53), over the same 16,486 agents.

The ladder is therefore a positive control and an upper bound on what any direction channel through this branch could buy *if it knew the future*, and (1) says how much of that bound is precision rather than selection. An observation-time cue buys selection only to the extent that it predicts the realised mode beyond what the model's existing inputs already predict. The oracle's advantage is that it does not predict the future mode but reads it.

A null at low power looks like a threshold. We first ran this ladder at 579 scenes, where it reads as a clean detection floor: significant at 10.0° (−6.0%, p=0.04), indistinguishable from zero at 15.1° (−1.4%, p=0.47), flat above. At 25k that second rung is −13.6%. The 579 ladder could never have seen it: scaling its own θ=0 amplitude (−15.8%) by the 25k decay predicts ≈6.2% there, against that rung's MDE of 6.7 pp. Its apparent floor was the angle at which a three-seed experiment went blind, and a threshold read off the first non-significant rung is a statement about power wearing the clothes of a property of the task. We report both ladders in Table IV for that reason.

TABLE IV

THE ORACLE LADDER. A DIRECTION-ONLY ORACLE CORRUPTED BY A FIXED PER-AGENT ANGULAR ERROR, EACH RUNG LABELLED BY THE MEDIAN BEARING ERROR ACTUALLY DELIVERED. TESTS ARE PAIRED t AGAINST EACH SEED'S OWN NO-POSE RUN, AND NEGATIVE FAVOURS THE ORACLE. MDE IS THE MINIMUM EFFECT A RUNG COULD DETECT AT 80% POWER GIVEN ITS REALISED SPREAD. AT 25K SCENES EVERY RUNG CONVERTS, INCLUDING 15.1°, WHICH IS WORSE AT EVERY PERCENTILE THAN THE BOX HEADING THE MODEL ALREADY RECEIVES. THE LADDER THEREFORE BOUNDS WHAT A DIRECTION CHANNEL COULD BUY RATHER THAN SETTING AN ACCURACY A CUE MUST REACH (SEC. V-A). THE SAME LADDER AT 579 SCENES REPORTS A FLOOR AT 15.1° THAT IS ITS OWN MDE: PROPORTIONALLY TO ITS θ=0 AMPLITUDE IT SHOULD HAVE READ ≈6.2% THERE, AND IT COULD RESOLVE 6.7.

Bearing error	n	ΔminADE	p	95% CI	MDE
25k scenes					
0.0°	3	−34.50%	0.0001	[−35.6, −33.4]	1.4
5.0°	3	−19.77%	0.0007	[−22.1, −17.5]	2.9
10.0°	3	−15.68%	0.0009	[−17.7, −13.6]	2.6
15.1°	3	−13.57%	0.0016	[−15.9, −11.2]	2.9
579 scenes: the same ladder, underpowered					
0.0°	3	−15.8%	0.021	[−25.6, −5.9]	12.3
5.0°	3	−7.6%	0.0004	[−8.2, −6.9]	0.8
10.0°	5	−6.0%	0.040	[−11.6, −0.5]	7.4
15.1°	5	−1.4%	0.472	[−6.5, +3.6]	6.7
24.9–60.1°	3	−0.1 to +2.7%	≥ 0.24	—	5.5–9.1

B. Why body facing does not pay

The oracle converts a bearing read off the realised future. Body facing is not that: it is an observation-time estimate of a quantity the model already measures. Three properties of it, on the population the ladder scores, say what is left for it to contribute.

It is a noisier copy of an existing input. The box heading estimates the future displacement bearing at 3.9° (+3 s) and 7.1° (+8 s) in median, while SMPL facing reaches 12.3° (n=5,824) and 15.1° (n=6,206). Box heading is the better estimate at every percentile we measured, the two differ by 10.0° in per-agent median, and both derive from the same returns. We calibrated the SMPL forward axis *against* the box heading before measuring either.

Most of its accuracy is the trajectory restated. 95.5% of the scored agents move, and for a mover the pipeline snaps root orientation to the direction of travel whenever the raw estimate deviates by more than 90° (Sec. III). Movers sit at 11.9°, while the 262 stationary agents, the only ones whose facing their velocity does not already imply, sit at 41.2°. The pooled median largely measures the snapping, and the distribution is heavy either way: 5.5° at the first quartile, 23.9° at the third, 45.9° at the ninetieth percentile.

Where it would be non-redundant, it is worst. On the one case where velocity carries no directional information (which way a stationary pedestrian steps off), SMPL facing predicts onset direction at roughly twice chance over 19,939 validation pedestrians: median error 44.2°, 50.6% within 45° (chance 90°, 25%). Informative, consistent with the 41.2° above, and not an artifact of our axis fitting: the box heading estimates the same quantity at 25.2°, and paired on the same events SMPL is 7.3° worse in median (Wilcoxon p=2.6×10^{-43}). Nor does pose take over where perception degrades: over 1,705 onset events the gap favours the box heading in all ten testable strata ($p < 10^{-5}$ each) and *widens* with range, to +32.9° at 41–75 m, because both derive from the same returns.

Facing is therefore redundant where it is accurate and inaccurate where it would be non-redundant, and neither half of that squeeze is the selection signal the oracle trades on. What survives it is what the content test measures directly: at 25k scenes +0.04%, with the data bounding it to ±1.7%, inside the ∼1.6% cost of the branch that would carry it.

Removing the redundant input does not rescue it. Redundancy says why little is left for facing to add, but

it is not by itself a mechanism that blocks conversion. So we tested the implied counterfactual: delete the box-heading channel, retrain all six cells, and repeat the content test, at the 579-scene rung and eight seeds. Nothing appears, anywhere. The difference in content effect between the two conditions is $+2.9$ pp on the 13,006 moving agents, *the wrong sign* for a benefit that redundancy had been masking, and -0.6 pp on the stationary ones (p=0.65). Seed-level intervals at this scale are wide (± 2 to ± 18 pp, widest exactly where minADE is largest), so we also average the eight runs per agent and test over agents, which cuts run noise without licensing a claim about retraining: that gives $[+2.4, +3.3]$ pp on movers, $[-1.5, +0.2]$ on the stationary subset, and -1.2 pp with $[-2.6, +0.2]$ in the top decile of turning, the stratum where neither velocity nor heading states the future bearing. The two analyses agree to within 0.15 pp on every stratum.

We can also say how much this ablation could ever have shown: deleting the heading channel costs the no-pose baseline only 1.5% on movers, because velocity stays in the input, so there was never much room for facing to fill. What remains unexcluded is the reading the content test's own bound asserts: that pose carries nothing this forecaster can use at any level of redundancy. At n=3 this comparison read -4.1 pp with all three seeds agreeing in sign: the third multi-percent effect here to dissolve under more seeds.

Nor does degrading it. The heading the baseline receives is an offboard label too (Sec. IV-E), so it may be better than a deployed tracker's, and facing might pay only where it is worse. Deleting the channel cannot test that: MTR's agent-centric frame, and the intention anchors defined in it, stay aligned to the clean heading. We instead corrupt the heading once, where that frame is set, with a fixed per-agent error $\mathcal{N}(0, \sigma^2)$ that frame, anchors, map and pose root all inherit, and repeat the content test at 25k scenes and three seeds (Table III). The pose root, rotated into the same corrupted frame, then states the correction. The corruption bites: the no-pose baseline loses 5.8% at σ=15° and 12.6% at 30°, where the median added error of 20.2° alone makes the heading a worse bearing estimate than body facing (12.3°). Pose recovers none of it: -0.14% (p=0.90) and -0.11% (p=0.93) against shuffled, and at 30° the interval caps recovery at 38% of the loss. A walker's track states its direction whatever the heading says, so we also score the 3,336 pedestrians stationary at prediction time, where it does not. There the corruption barely costs the baseline (-1.1% and $+4.3$%, both n.s.), since the labelled heading was already a weak cue for where a standing pedestrian goes (25.2° to the onset direction, above), and pose against shuffled reads $+1.2$% and $+1.3$% (p=0.049), if anything the wrong sign. Nor is the walking majority drowning out a signal there: fine-tuning each arm from its clean model for five epochs on standing pedestrians alone leaves pose against shuffled on them at -1.2% (p=0.28), no better than before ($+1.5$ pp, p=0.36). The clean rung reproduces our main cells to within 0.4%, and a second learning rate at 30° does no better ($+0.7$%, p=0.48). However the heading is degraded, facing stays redundant, and what a degraded track loses and the body restores is speed (Sec. IV-E).

C. The cheaper pathway converts a bearing, and still not pose

Early fusion removes the branch tax (Sec. IV-B), which makes it the pathway worth recommending, so we ran the same ladder through it, carrying the oracle in the per-timestep agent channels instead of the branch. At a given rung the six channels equal the branch's six inputs bit for bit, so the rungs keep their measured-error labels.

At 579 scenes this ladder reads as a dead pathway. No rung is significant, *including the uncorrupted one*: -4.5% at θ=0 (p=0.44), -1.0% at 5.0° (p=0.53) and $+0.2$% at 10.0° (p=0.94), and the per-seed pattern looks like a mechanism rather than noise. The side branch converts the uncorrupted channel on all three seeds (-16.5%, -19.3%, -11.5%) while early fusion converts it on one (-14.0%, against $+0.7$% and -0.3%), with the two failing seeds sitting at their no-pose level rather than scattering about it. A difference in *how many seeds move at all* does not look like something a wide interval explains away.

It was. The MDE at θ=0 is 25.5 pp, and we re-ran the rung at 25k scenes, where the branch ladder's MDE was 1.4–2.9 pp. Early fusion converts the uncorrupted channel there on *every* seed, by -31.2% (p=0.0002, 95% CI $[-32.9, -29.5]$), against the branch's -34.5% at the same rung and denominator, and the corrupted rungs follow, -19.4% at 5.0° and -15.5% at 10.0° ($p \leq 0.001$), each within 0.4 pp of the branch. So the 579 reading was the same power artifact twice over: once in the significance column, and once in a per-seed signature we trusted because it was not a p-value. Three runs at 579 scenes cannot tell a pathway that ignores a channel from one that converts it by a third.

This also settles the free-carriage result of Sec. IV-B the other way. "No tax, no benefit" admitted two readings: that early fusion integrates pose at no capacity cost, or that it largely ignores the appended channels, so that there is nothing to cost and nothing to gain. A perfect channel separates them, and at the scale where we can resolve the difference the first reading is correct. Early fusion reads what it is given. Pose through that pathway is therefore a cue that arrives, is used when it carries something, and still buys nothing, which removes the last place the null could have been an artifact of integration rather than a property of the cue.

D. The complementary cue is barely measured

Gaze precedes locomotion, and head orientation is an established intent cue for vehicle-based pedestrian path prediction [19], [20], so head yaw relative to the torso is a plausible *independent* cue. It is very nearly not recoverable here. In the fitted meshes head-versus-torso yaw has a standard deviation of 3.8°, while against the ground-truth 3D keypoints released with Waymo Perception [21], [22] (n=8,234) the real quantity varies 19.2°: a fivefold under-dispersion, so the estimator is largely emitting its prior. Consistently, head yaw carries no usable signal for onset direction ($r = -0.03$, n.s.), and the head is the one joint in Table V whose interval includes zero.

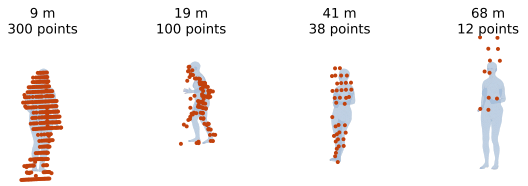


Fig. 3. What the pose estimator has to work with. Raw per-pedestrian LiDAR returns (orange) and the SMPL mesh fitted to them (blue), at four ranges and a common metric scale. Each panel is a frame whose return count equals the median for its range band over the validation segments: 300, 100, 38 and 12 points. Past ≈40 m a pedestrian is a few dozen returns, mostly on head and torso, while the mesh over them still has fully articulated limbs: at that range the articulation is largely the AMASS prior. That applies to the far panels only. Scored against ground-truth keypoints across all ranges, every limb joint carries real skill, and it is the *head* the pipeline emits from its prior (Table V). Sparsity at range is therefore one contributor to the facing error of Sec. V-B, where the pose-versus-heading gap widens to $+32.9°$ at 41–75 m, not its whole explanation.

TABLE V

PER-JOINT RECOVERABILITY AGAINST GROUND-TRUTH KEYPOINTS. MP IS THE ERROR OF ALWAYS PREDICTING THE MEAN POSE (EQUAL TO THE JOINT'S GROUND-TRUTH SPREAD BY CONSTRUCTION), AND $\mathrm{SKILL} = 1 - \mathrm{MPJPE}/\mathrm{MP}$. THE HEAD'S MPJPE BEATS BOTH WRISTS WHILE CARRYING A TENTH OF THEIR SKILL. INTERVALS RESAMPLE PEDESTRIANS (n=534), NOT FRAMES, SINCE FRAMES WITHIN A TRACK ARE STRONGLY CORRELATED. EVERY LIMB INTERVAL CLEARS ZERO BY A WIDE MARGIN (LOWEST BOUND $+0.43$). THE HEAD'S IS $[-0.003, +0.089]$, THE ONE INTERVAL THAT CONTAINS IT.

Joint	MPJPE (mm)	MP (mm)	SKILL
Elbows	91.2 / 91.5	252.9 / 248.9	+0.64 / +0.63
Hips	39.8	97.9	+0.59
Wrists	118.3 / 125.4	271.1 / 276.4	+0.56 / +0.55
Shoulders	87.3 / 89.0	171.5 / 164.0	+0.49 / +0.46
Ankles	114.0 / 116.6	221.6 / 223.8	+0.49 / +0.48
Knees	78.4 / 81.5	150.8 / 151.2	+0.48 / +0.46
Head	**111.6**	**116.9**	**+0.045**

At 20–40 m the head is a near-spherical handful of points with little geometry to constrain yaw.

E. MPJPE is the wrong acceptance metric

A single mean error cannot distinguish an estimator that tracks a joint imprecisely from one that emits its prior. We therefore score each joint against the strongest trivial competitor, always predicting the mean root-aligned pose: $\mathrm{SKILL}_j = 1 - \mathrm{MPJPE}_j/\mathrm{MP}_j$, where values at or below zero mean the estimator contributes nothing at that joint.

Evaluated on Waymo Perception against ground-truth keypoints (1,789 frames, 534 pedestrians), every limb joint scores $+0.457$ to $+0.639$ (Table V). The head scores $+0.045$, yet its MPJPE (111.6 mm) is *better* than either wrist (118.3/125.4 mm). Its error is flattering not because it is well estimated but because it barely improves on its own mean-pose baseline (116.9 mm), whereas the hips, whose ground-truth spread is *smaller* at 97.9 mm, are predicted at 39.8 mm. Across the thirteen joints MPJPE and skill are essentially uncorrelated (Spearman $\rho = -0.11$, p=0.72): the aggregate is not merely a noisy proxy for information content but carries almost none of it, and for the head it actively inverts the ordering. We recommend reporting per-joint skill alongside it whenever a pose estimator is accepted into a downstream stack.

F. A general condition

Two results locate where pose could have helped. At 2 s, minADE is 0.091 m, and a perfect future-direction oracle through the same branch removes only 1.4 cm of it (-14.7%): relative headroom exists at short horizons, but in absolute terms there is almost nothing for any cue to win. This is a property of the forecaster's own residual, not of constant velocity: single-mode constant-velocity extrapolation [23] gives 0.310 m at 2 s on the same set, so short-horizon forecasting is *not* CV-saturated (a comparison that favours the model, since minADE takes the best of K=6 modes). Saturation here means the baseline's own remaining error is a few percent of the displacement. At long horizons the residual is dominated by destination and intent, which body configuration does not disclose, and stratifying by how badly constant velocity fails still yields no effect.

In general form, and offered as a hypothesis from one modality on one dataset rather than a law: *an auxiliary modality improves a forecast only where the baseline's own residual leaves room in absolute terms, where that residual is not dominated by intent the modality cannot observe, and where the modality carries information about the cue that the model's existing inputs do not already carry.* All three clauses must hold. The third is not a comparison of marginal accuracies, and this paper is the reason to state it that way: Sec. V-A converts a channel less accurate at every percentile than an input the model already has, so being the noisier estimate is not by itself disqualifying. What disqualifies facing is that its error is *shared* with the input it would supplement. It is calibrated against the box heading, snapped to the direction of travel for the 95.5% of agents that move, and recovered from the same returns, so where it is accurate it is redundant and where it would be independent it is worst (Sec. V-B). Sec. IV-E tests the clause from the other side: leave the body as it is, degrade the input it duplicates, and the same body pays. Redundancy is therefore a property of a modality together with the inputs beside it, and an offboard-labelled benchmark sets those inputs at offboard quality. The check this licenses is still cheap and still skipped, and it is still a measurement on the dataset with no training: how much of the target does the candidate cue explain that the existing inputs do not?

VI. WHAT THIS MEANS IN PRACTICE

For an AV stack, four consequences follow. First, before integrating an auxiliary modality, measure not only its error on the cue it is meant to carry but how much of that cue it explains *beyond the inputs the model already receives*, and take those inputs at the quality a vehicle will have. That costs no training. On offboard labels it predicts our null, and a simulated causal tracker, a few lines in a data loader, shows what the same body is worth on a realistic track. Second, an oracle ladder is still worth its handful of runs, as a positive control and as

an upper bound on the pathway, but size it so its MDE is below the effect you would act on, and do not read its angular axis as an acceptance threshold: ours converts a channel less accurate than an input the model already has, because an oracle knows which future occurred and no sensor does (Sec. V-A). Third, the capacity cost of an extra branch is real ($\sim$1.6% here, at the best-epoch operating point) and is easily mistaken for a modality effect in either direction. Early fusion avoids it while still reading the channel, as an oracle through that pathway demonstrates, and a shuffled-pose control separates a benefit from a capacity effect either way. Fourth, do not accept a pose estimator on aggregate MPJPE alone: check per-joint skill, because the joints an aggregate flatters may be the ones carrying a prior.

Limitations. The largest rung we train is 25k scenes of the 299,220 exported, so the full rung is a compute limit rather than an annotation one. At eight seeds the minimum detectable content effect is $\approx$1.25% at 80% power, and the ladders run at n=3 per rung (Tables IV and III). We test one forecaster, one dataset and one pose estimator. We ladder two integration pathways rather than one, but two is not every way a cue can enter a model, and the early-fusion ladder stops at 10.0° (Sec. V-C). The onboard track is simulated: Gaussian detection noise through a constant-velocity Kalman filter rather than a detector's errors. Its bodies are still fitted inside offboard boxes, and we have not run it through early fusion. The redundancy counterfactual, which deletes the heading channel to see whether facing then pays, is run at the 579-scene rung rather than at 25k, and its seed-level intervals there are wide enough that the agent-level analysis carries the turn strata. WOMD ships camera data only as quantised tokens, so a camera-based pose or gaze estimate cannot be tested at this scale. On Waymo Perception, where raw imagery exists, an RGB estimator (HMR2.0) reaches 106.2 mm root-aligned MPJPE against LiDAR-HMR's 87.6 mm on the same 347 frames, so the obvious better-pose route is the worse one here. We do not model pose–pose interaction.

VII. Conclusion

Body pose, recovered from LiDAR, does not improve pedestrian trajectory forecasting on WOMD's offboard tracks, and it does once those tracks are replaced by what a causal onboard tracker would report. On the offboard labels the null is controlled against a shuffled-pose baseline and holds across scale, representation, integration and box-heading quality, and an oracle through the same branch rules out a dead pathway. Under a simulated tracker own pose beats a stranger's by 6.0%, on moving pedestrians, where gait carries the speed the tracker loses. The two results are one mechanism: a cue pays only for what it carries beyond the model's existing inputs, and an offboard track leaves the body nothing to add. Degrading the box heading does not make body facing pay. As a bearing it is a noisier copy of that heading, and worst, at 41.2°, on the stationary pedestrians it was meant to serve. The oracle ladder shows what the pathway would convert, a direction channel at every accuracy we trained, and that what it sells beyond precision is knowledge of which future occurred, which no sensor supplies. The lesson generalises past pose. Before integrating a modality, measure how much of the cue it carries beyond the model's existing inputs, take those inputs at the quality a vehicle will have rather than from offboard labels, and make sure any ladder you judge it against can resolve the effect you would act on.

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